# Similarity by Selected Traits

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

## Teach a distance as a declared comparison of represented features.

Supply all vectors, code and axis meanings, and keep independent targets separate.

**Keep one habit:** Require coordinate gaps, sum, tie policy and suitability limits beside every nearest-match claim.

MAIN LESSON · SECTION 1

## Prepare this connected week

> Use one main plan and keep practice separate from the independent first attempt.

### Learning goal and precision

Primary target: Represent items with numeric features

Teach Week 22 as a two-session investigation using fictional data. A fictional activity chooser represents each activity with energy and group-work ratings on a 0 to 5 **scale**. A learner requests [2, 4].

We compare options by adding absolute differences. This is a chosen rule for **similarity**; it is not a **learned model** or an objective measure of quality.

Assess the three stated outcomes through the independent task, not by copying the worked result.

By the end, I can calculate a feature **distance** and state how scaling changes its meaning.

## What to prepare

- Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.
- Python 3 editor with a fresh local practice file
- Paper trace table and the student workbook
- Fictional **input** records from this lesson; no real student identifiers

## Main teaching plan

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

## Opening retrieval and readiness

Without opening the old lesson, explain one key idea from Repair Bias in Data. Give an example and a mistake that the idea helps you avoid.

Look for: Identify selection and coverage bias

Reference example, not the only acceptable wording: Afternoon or evening visitors, and people unable to attend in the morning.

Without opening the old lesson, explain one key idea from Missing Data. Give an example and a mistake that the idea helps you avoid.

Look for: Detect absent and unusable values

Reference example, not the only acceptable wording: 0 is a recorded allowed duration. The other two are unknown under this schema.

Without opening the old lesson, explain one key idea from Choose Informative Questions. Give an example and a mistake that the idea helps you avoid.

Look for: Fit a root question and leaf **prediction**s from **training** examples

Reference example, not the only acceptable wording: The root asks near, with branches 0 and 1. Leaves predict clear for 0 and blocked for 1. Depth is one.

**Training** labels determined both leaf majorities and the error comparison that selected near rather than tall.

The absolute difference is 4.

## ▼ Earlier wording, alternatives and picture transcripts

### Teach the learning cycle

### Retrieval answer guidance

### Repair Bias in Data

### Missing Data

### Choose Informative Questions

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

### What the learner must demonstrate

Represent books as [animal,short,illustrated] with 1=yes. Query=[1,1,0]; A=[1,0,0], B=[1,1,1], C=[0,0,0]. Count mismatches. What changes if illustrations matter twice as much?

### Concept to explain

Week 21: **Similarity** by Selected Traits

| Term | Meaning |
|---|---|
| similarity | How alike two items are according to selected features and a declared comparison method. |
| distance | A specified measure of how far apart two representations are. |
| coordinate | A value identifying position along an agreed axis or within a space. |
| scale | The units and range used to represent or compare a quantity. |

### Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

### Preparation for the connected explanation

### Readiness answer

### End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

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MAIN LESSON · SECTION 2

# Teach distance through coordinate gaps and explicit ties

> Keep activity and book targets distinct.

## Define the activity representation

A fictional activity chooser represents each activity with energy and group-work ratings on a 0 to 5 **scale**. A learner requests [2, 4]. We compare options by adding absolute differences.

This is a chosen rule for **similarity**; it is not a **learned model** or an objective measure of quality.

### Predict and execute its complete script

```
target = [2, 4]
options = {"Sketch": [1, 3], "Relay": [5, 4]}
for name in options:
    features = options[name]
```

```
    distance = (abs(features[0] - target[0])
                + abs(features[1] - target[1]))
    print(name, distance)
```

## Explain feature order and scale meaning

abs gives a nonnegative gap. Under this rule Sketch is closer. Feature order must match: first energy, then group work. Both **scale**s are 0 to 5, but that does not prove one unit has identical practical meaning in both features.

## Collect the two-distance exercise

> **Find the distance from target [2, 4] to [2, 4] and to [3, 3].**

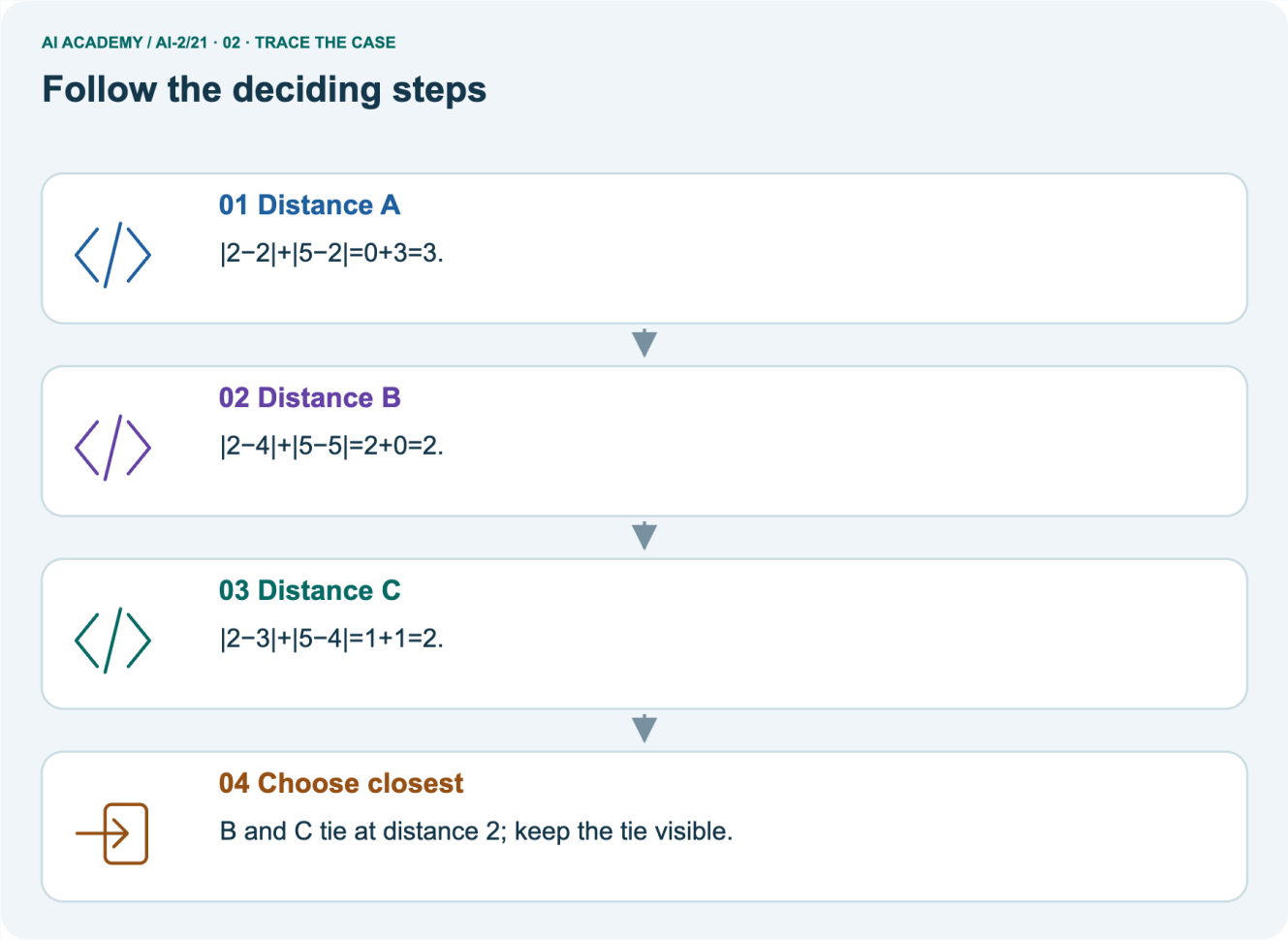
## Introduce the separate book target and all-ties rule

A fictional toy book matcher represents two preferences with numeric features.

Target vector [2,5]. Candidate A=[2,2], B=[4,5], C=[3,4].

Use **Manhattan distance**: add absolute **coordinate** differences; return all smallest-**distance** candidates.

## Trace the tie and representation limits




Follow the labelled picture with the matching explanation. Open the vector diagram

Each **coordinate** contributes according to the declared **distance** rule. Smaller **distance** means closer in this representation, not automatically a better reading experience.

If one **coordinate** uses a much larger numerical **scale**, it can dominate the sum even when it is not more important. Feature units and scaling belong in the method record.

A toy numeric encoding does not establish meaningful interpersonal **distance**s or suitability of a recommendation.

## Change the weight, not the vectors

Change: Give the first-**coordinate** difference weight 2 while the second keeps weight 1.

**Prediction**: **Distance**s become A=3, B=4, C=3; A and C tie.

Reason: The representation’s comparison rule changes; the supplied vectors stay the same.

## Elicit the picture explanation

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

### ▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

**Input** → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. Reader target: length=2, difficulty=3.

2. Book A: (2,2) → grid **distance** |2−2|+|2−3|=1.

3. Book B: (4,3) → **distance** |4−2|+|3−3|=2.

4. A is closer under these features; changing features can change the answer.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

#### Concrete teaching case

Traits are [animals,adventure]. A=[1,1], B=[1,0], C=[0,0]; query=[1,0].

Use number of differing traits as **distance**.

### Exact explanation to model

**Distance**s: A=1, B=0, C=1. B is nearest with equal feature weights.

### Explain the added worked example

### Key terms

| Term | Meaning |
|---|---|
| Feature vector | An ordered list of numeric features |
| Distance | A number describing difference under a chosen rule |
| Manhattan distance | The sum of absolute feature differences |
| Normalization | Rescaling features to a comparable scale |

## Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

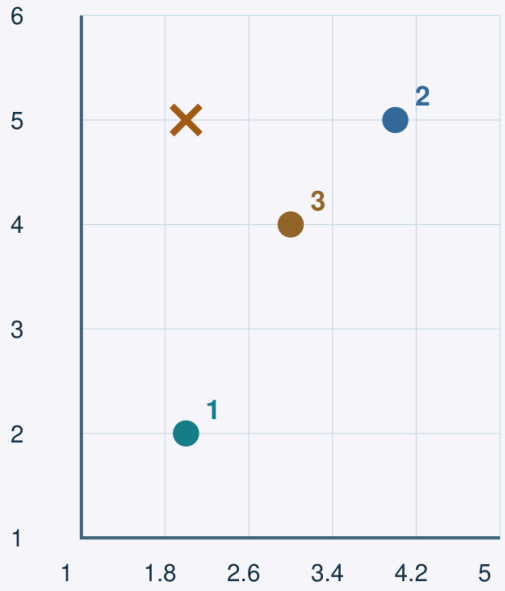
| Thinking move | Teacher prompt |
|---|---|
| Plan | What is given? What is the question? Which procedure fits these facts? |
| Do | Point to each input and intermediate step in the worked trace. Name the rule that allows the next step. |
| Check | Does the output answer the question? Which assumption or limitation prevents a larger claim? |

## A pictorial worked case: Similarity by Selected Traits

### The facts we will use

AI ACADEMY / AI-2/21

**Toy feature positions**




Horizontal: first toy feature; vertical: second toy feature

1: A at (2, 2)

2: B at (4, 5)

3: C at (3, 4)

Cross: Target at (2, 5)

Distance rule: Manhattan distance; horizontal=first feature, vertical=second feature

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/21

# Similarity by Selected Traits

**Start with the supplied facts**

Follow each arrow. Ask which fact allows the next step.

### 1. Distance A

$|2-2|+|5-2|=0+3=3$.

### 2. Distance B

$|2-4|+|5-5|=2+0=2$.

### 3. Distance C

$|2-3|+|5-4|=1+1=2$.

### 4. Choose closest

B and C tie at distance 2; keep the tie visible.

**CHANGE ONE THING**

Give the first-coordinate difference weight 2 while the second keeps weight 1.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

### Read the picture one step at a time

| STEP | WHAT HAPPENS AND WHY |
| --- | --- |
| 1. Distance A | $\lvert 2-2\rvert+\lvert 5-2\rvert=0+3=3$. |
| 2. Distance B | $\lvert 2-4\rvert+\lvert 5-5\rvert=2+0=2$. |
| 3. Distance C | $\lvert 2-3\rvert+\lvert 5-4\rvert=1+1=2$. |
| 4. Choose closest | B and C tie at distance 2; keep the tie visible. |

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

**Why this result follows**

**What this example does not establish**

**Change one thing in the pictured case**

**Ask while showing the picture**

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MAIN LESSON · SECTION 3

## Guide the reasoning, then remove help

> Collect a response from each learner before discussing the worked key.

### Prompt before revealing

Completion task: From target [1, 2], find **distance**s to A [2, 2] and B [0, 4].

## Expected reasoning and fading help

Teacher check: A: 1 + 0 = 1; B: 1 + 2 = 3.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

### Check the misconception before discussion

Use the worked case for **Distance** between **feature vector**s. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: **Similarity** depends on the features collected and the **distance** rule. Missing accessibility, cost or preference information can make the closest item unsuitable.

Hard requirements should be checked separately.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

## Use the pictorial check

Explain why B ceases to be a closest option without any candidate data changing.

Its first-**coordinate** gap 2 now contributes 4, raising its **distance** above A and C’s 3.

Changed-case reasoning: **Distance**s become A=3, B=4, C=3; A and C tie. The representation's comparison rule changes; the supplied vectors stay the same.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

## Keep each learner explaining

- Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.
- Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.
- In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

### ▼ Earlier wording, alternatives and picture transcripts

Demonstrate and guide practice

#### Board sequence

Write the given facts: Traits are [animals,adventure]. A=[1,1], B=[1,0], C=[0,0]; query=[1,0].

Write the question to explain: Use number of differing traits as **distance**.

Reveal after **prediction**s: **Distance**s: A=1, B=0, C=1. B is nearest with equal feature weights.

Use the earlier changed question as optional later practice. Reserve Rank independently for the first independent attempt before hints or feedback.

#### Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

### Model these guided questions

**Distance**s 0 and 2. Zero means identical recorded features, not identical activities in every way. Exit: No. Only the recorded features match; unrecorded needs and activity details may differ.

### Case-specific teaching sequence

| Stage | Teacher action |
|---|---|
| Session 1 start to 7 minutes | Retrieve the prerequisite and explain the purpose in the opening section. |
| 7 to 20 minutes | Read the worked example and diagram; ask learners to trace each step aloud. |
| 20 to 33 minutes | Complete W1 and W2, then compare predictions with observed results. |
| 33 to 45 minutes | Collect the guided prediction before running; discuss a mismatch and its cause. |
| Session 2 start to 8 minutes | Retrieve the main rule without showing the previous answer. |
| 8 to 20 minutes | Use W3 to diagnose and repair the stated failure; rerun an earlier case. |
| 20 to 35 minutes | Collect W4 independently, including code or plan, predictions and explanations, before hints. |
| 35 to 45 minutes | Discuss W5 and the exit; use the fresh retry when support was needed. |

### Reduce help deliberately

### Individual reasoning check

### Pictorial practice question

### Reasoned teacher answer

# Run the supplied investigation

> Keep expected results, observed paper traces and remaining limits distinguishable.

## Supplied materials and answer guidance

Activity distances Sketch2,Relay3; target to itself0 and to[3,3]2. Three-book original distances A3,B2,C2; weight(2,1) gives A3,B4,C3. Six-book lab extends the same three book vectors: equal A3,B2,C2,D7,E8,F3, closestB/C; weighted A3,B4,C3,D9,E11,F6, closestA/C. Keep the rule and target beside each calculation. No learned model or real suitability result is supplied. Reserve target[3,2] for W4.

## Run and record

1. Predict → Test → Record → Explain
2. Keep expected and actual results separate. Surprises are useful data when recorded honestly.
3. Use the six supplied toy book vectors and target[2,5]. Label both grid axes, calculate equal-weight **distance**s, then change only the pace-gap weight to2. Keep all ties visible and distinguish changed rules from changed data.
4. What did the hardest test teach you that the normal test did not?

## Keep an inspectable record

| Record | What to capture |
| --- | --- |
| Input or evidence | Exact data, instruction or claim |
| Expected result | Answer or safe fallback declared before the run |
| Observed result | Actual result, including errors |

| Record | What to capture |
| --- | --- |
| Explanation | Rule, calculation or evidence explaining the outcome |
| Next test | One justified change and a new prediction |

## Diagnose the first misunderstanding

A successful run proves the conclusion: **Similarity** depends on the features collected and the **distance** rule. Missing accessibility, cost or preference information can make the closest item unsuitable.

Hard requirements should be checked separately.

Matching the example is enough: Require the new W4 case, explicit **prediction**s and an explanation. Fresh retry: target [2, 3]; X [0, 3], Y [2, 2], Z [3, 4] have **distance**s 2, 1, 2. Y wins and X/Z tie.

A convenient shortcut preserves meaning: If one feature is duration in minutes up to 60 and another a rating up to 5, raw **distance** may be dominated by duration.

Specify scaling or weights before ranking; do not silently change units to get a preferred winner.

## Choose support from the response

| What the response shows | Next teaching action |
| --- | --- |
| Missing or misread facts | Read the case aloud, mark supplied versus unknown facts, and let the learner restate it. |
| Wrong procedure | Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation. |

| What the response shows | Next teaching action |
| --- | --- |
| Arithmetic or implementation error | Trace intermediate values or lines; preserve the first mismatch and retry that step. |
| Correct result without a reason | Ask for a trace and a changed case before marking independent understanding. |
| An unsupported wider claim | Ask which supplied evidence supports the claim and which further observation would be needed. |

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

### ▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

#### Make the lab runnable

Use the supplied investigation for shared practice. Keep predictions and observed paper-trace results separate. Reserve Rank independently for the individual first attempt before feedback; do not use the reserved case as a paired lab task.

#### Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

#### Address these additional misconceptions

#### Match the support to the first error

MAIN LESSON · SECTION 5

# Assess the fresh distance comparison

> Keep the independent key until after the first attempt.

### Selected question and checked key

Selected case: Target is [3, 2]. Options A [1, 2], B [3, 4], C [4, 2]. Calculate **distance**s, select the closest, and identify a tie among the others.

**Distance**s A 2, B 2, C 1. C is closest; A and B tie. Do not claim a unique ranking of the tied pair without a rule.

## Judge gaps, totals and limits

| Dimension | Evidence required |
|---|---|
| Explanation | Names the applicable idea or procedure and ties it to the supplied facts. |
| Trace and result | Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution. |
| Limits | States the assumptions, missing evidence or conditions under which the result would change. |

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

## Give a repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

## W1 feature-order key

For [4, 1] under the worked feature order, identify energy and group-work ratings.

Energy 4, group work 1.

Use the declared feature order: position 0 means energy and position 1 means group work. The vector [4,1] therefore records energy 4 and group work 1.

Reversing the interpretation would change the meaning and could change the **distance** calculation.

## W2 distance key

From target [1, 2], find **distance**s to A [2, 2] and B [0, 4].

A: 1 + 0 = 1; B: 1 + 2 = 3.

## W3 schema-repair key

One record uses [group work, energy] while others use [energy, group work]. Why is the **distance** unreliable?

It compares different meanings in each position. Reorder using the documented schema before computing.

## W5 similarity limit

Two activities have **distance** zero. Must they suit every learner equally well?

No. Only the recorded features match; unrecorded needs and activity details may differ.

### ▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

### Worked case answer

**Distance**s: A=1, B=0, C=1. B is nearest with equal feature weights.

### Additional transfer practice question

Score **similarity** using one point for same topic and one for same difficulty. Target is space, easy. A is space, hard; B is animals, easy. What happens?

### Reasoned practice answer

Each scores one point. The chosen traits create a tie; use a stated tie rule or request another relevant preference.

### What counts as evidence

| Criterion | 0 points | 1 point | 2 points |
|---|---|---|---|
| Result | Incorrect or absent | Partly correct | Correct with necessary conditions |
| Reasoning | No supporting trace | Some steps | Clear mechanism, calculation or source support |
| Limit | Overclaims certainty | Vague limitation | Relevant uncertainty or missing fact |
| Transfer | Copies without adapting | Adapts with prompts | Solves a changed case independently |

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

### Feedback

Ask the learner to find the **evidence** that challenges this claim: "**Similarity** always gives exactly one winner." Keep the first attempt and an uncoached retry separate.

### Original transfer answer

Equal-weight **distance**s are A=1, B=1, C=2. Doubling the illustration mismatch makes A=1, B=2, C=2, so A becomes nearest. Weight choices express a task preference; they are not objective truth about book quality.

**Explanations for the additional workbook examples**

**W1 Read a vector**

**W2 Compute distances**

**W3 Repair feature order**

**W4 Rank independently**

Target is [3, 2]. Options A [1, 2], B [3, 4], C [4, 2]. Calculate **distance**s, select the closest, and identify a tie among the others.

**Distance**s A 2, B 2, C 1. C is closest; A and B tie. Do not claim a unique ranking of the tied pair without a rule.

**W5 Limit similarity**

**Assess independent transfer**

**Selected case answer**

**Feedback that the learner uses**

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MAIN LESSON · SECTION 6

# Connect, adapt and return

> Retain the project evidence and revisit the reasoning on a changed case.

## Project evidence to keep

Make selected traits and their **scale**s visible in every recommendation explanation.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Specify recommender feature meanings, units, scaling and tie policy before explaining nearest matches.

## Support while preserving the reasoning goal

Trace one record or condition at a time with counters and labelled columns. Stretch: introduce a tie, boundary or missing field and declare its handling policy before rerunning.

Use a small square grid and count horizontal plus vertical steps. Extension: re**scale** a minutes feature by its declared range and compare rankings transparently.

Independent W4 **evidence**: **Distance**s A 2, B 2, C 1. C is closest; A and B tie. Do not claim a unique ranking of the tied pair without a rule. Record whether the learner needed no help, a hint or explanation.

Fresh retry: target [2, 3]; X [0, 3], Y [2, 2], Z [3, 4] have **distance**s 2, 1, 2. Y wins and X/Z tie.

## Plan the delayed check

Suggested return points in this level: ai-2/22, ai-2/24, ai-2/27. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

| Record | Teacher evidence |
| --- | --- |
| Learner code and date | |
| Concept and case used; prior exposure | |
| Explanation / trace / limits | |
| Support and first repair | |
| Changed case and delayed outcome | |
| Next teaching action | |

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

## Bridge to next week

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 22: Nearest-Neighbor Reasoning. Retrieve today's **evidence** and identify what carries forward and what is new.

Next is Week22: Nearest Neighbor Reasoning. Carry forward the declared representation, distance, candidate coverage and tie policy before using a neighbour's label to predict a new case.


▼ **Earlier wording, alternatives and picture transcripts**

Connect and extend learning

**Transparent Book Recommender**

**Support and stretch**

**Exit reflection**

**Bridge to the next week**

**Support and fresh reassessment**

**Plan the delayed return**

**Project evidence from this case**

# Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference: alternative planning checklist**

REFERENCE · SECTION 7

## Reference: alternative planning checklist

> Use as a reminder alongside the main route rather than teaching the cycle twice.

### Preparation and schedule

Teaching target: Represent items with numeric features

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
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> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

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> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

## Practice and assessment reminders

Before class, solve the supported practice: From target [1, 2], find **distance**s to A [2, 2] and B [0, 4].

Reasoned practice answer: A: 1 + 0 = 1; B: 1 + 2 = 3.

Check every learner before discussion: Use the worked case for **Distance** between **feature vector**s. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: **Similarity** depends on the features collected and the **distance** rule. Missing accessibility, cost or preference information can make the closest item unsuitable.

Hard requirements should be checked separately.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Give the first-**coordinate** difference weight 2 while the second keeps weight 1. Require a **prediction** and a reason before comparing. Expected result: **Distance**s become A=3, B=4, C=3; A and C tie.

Reason: The representation’s comparison rule changes; the supplied vectors stay the same.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

| Criterion | Not yet | With support | Independent |
|---|---|---|---|
| Explanation | Names an answer without a deciding fact | Uses a hint to link fact and decision | Explains the deciding fact in own words |
| Trace or test | Cannot connect input and result | Completes steps with prompts | Traces or tests a changed case accurately |
| Limits and revision | Claims more than evidence supports | Names a limit after a prompt | States a defensible limit and useful next test |

Project checkpoint: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference: feedback and records**

REFERENCE · SECTION 8

# Reference: feedback and records

> Use digital tools where they help retain an explanation and its evidence.

## Keep practice and independent evidence separate

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

### Repair and portfolio records

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.
</details>

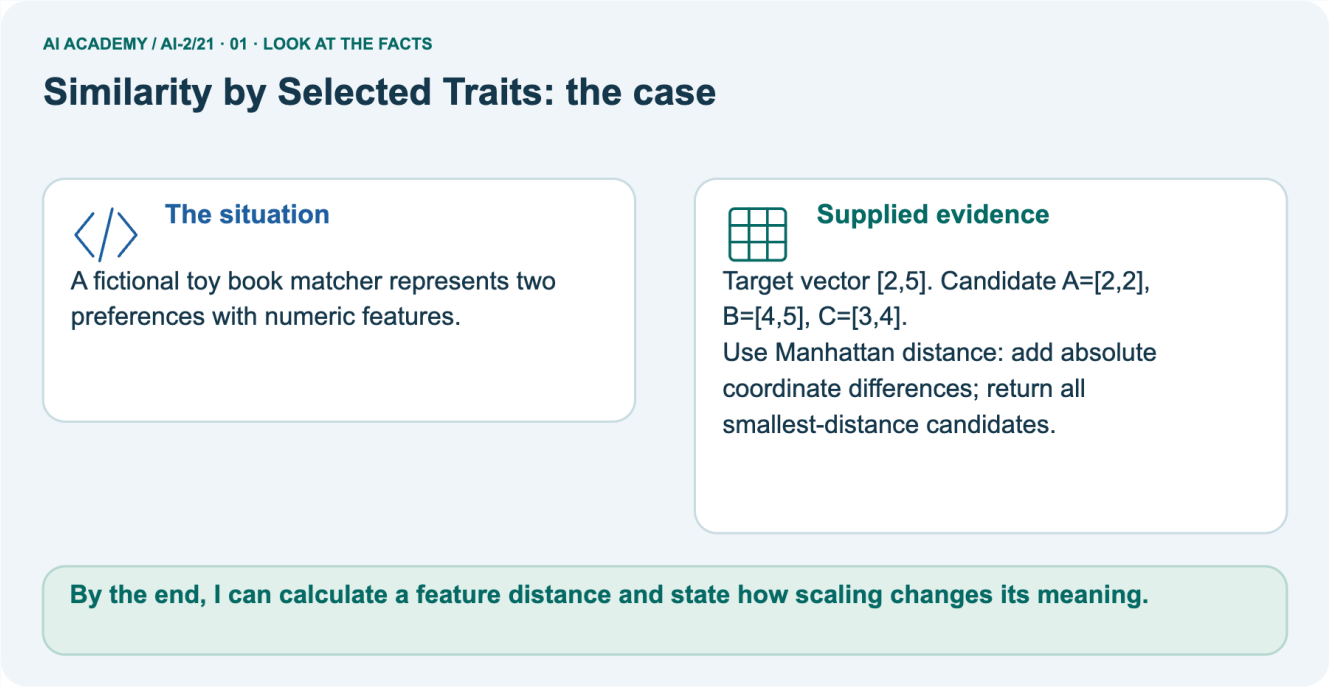
<details>
<summary>Reference: teach with the six pictures</summary>
</details>

REFERENCE · SECTION 9

## Reference: teach with the six pictures

> Choose the picture for the teaching move and connect it to the explanation.

## 01. Similarity by Selected Traits: the case




Follow the labelled picture with the matching explanation. Open the vector diagram

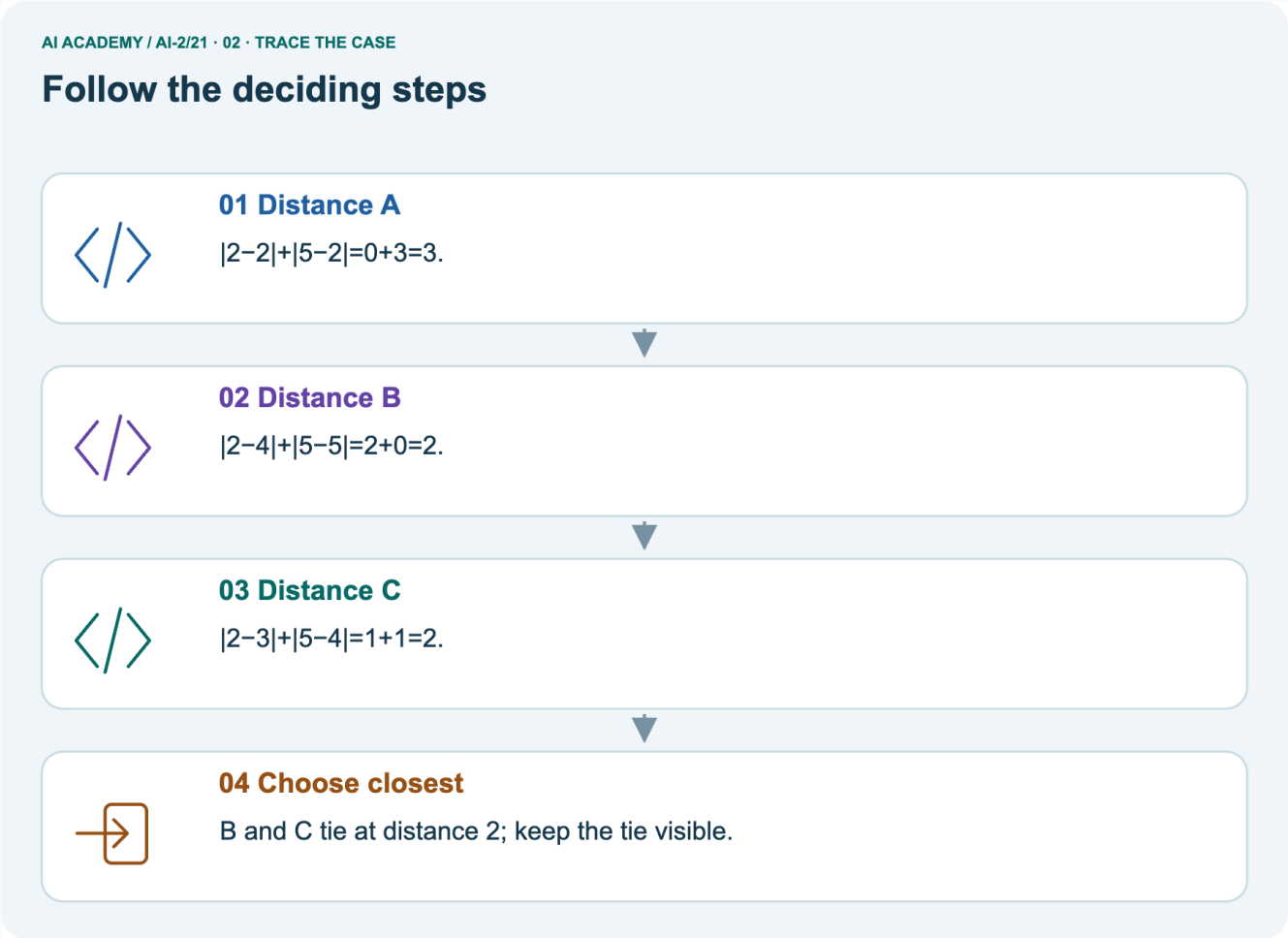
A fictional toy book matcher represents two preferences with numeric features. Supplied facts: Target vector [2,5]. Candidate A=[2,2], B=[4,5], C=[3,4].

Use **Manhattan distance**: add absolute **coordinate** differences; return all smallest-**distance** candidates.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

## 02. Follow the deciding steps



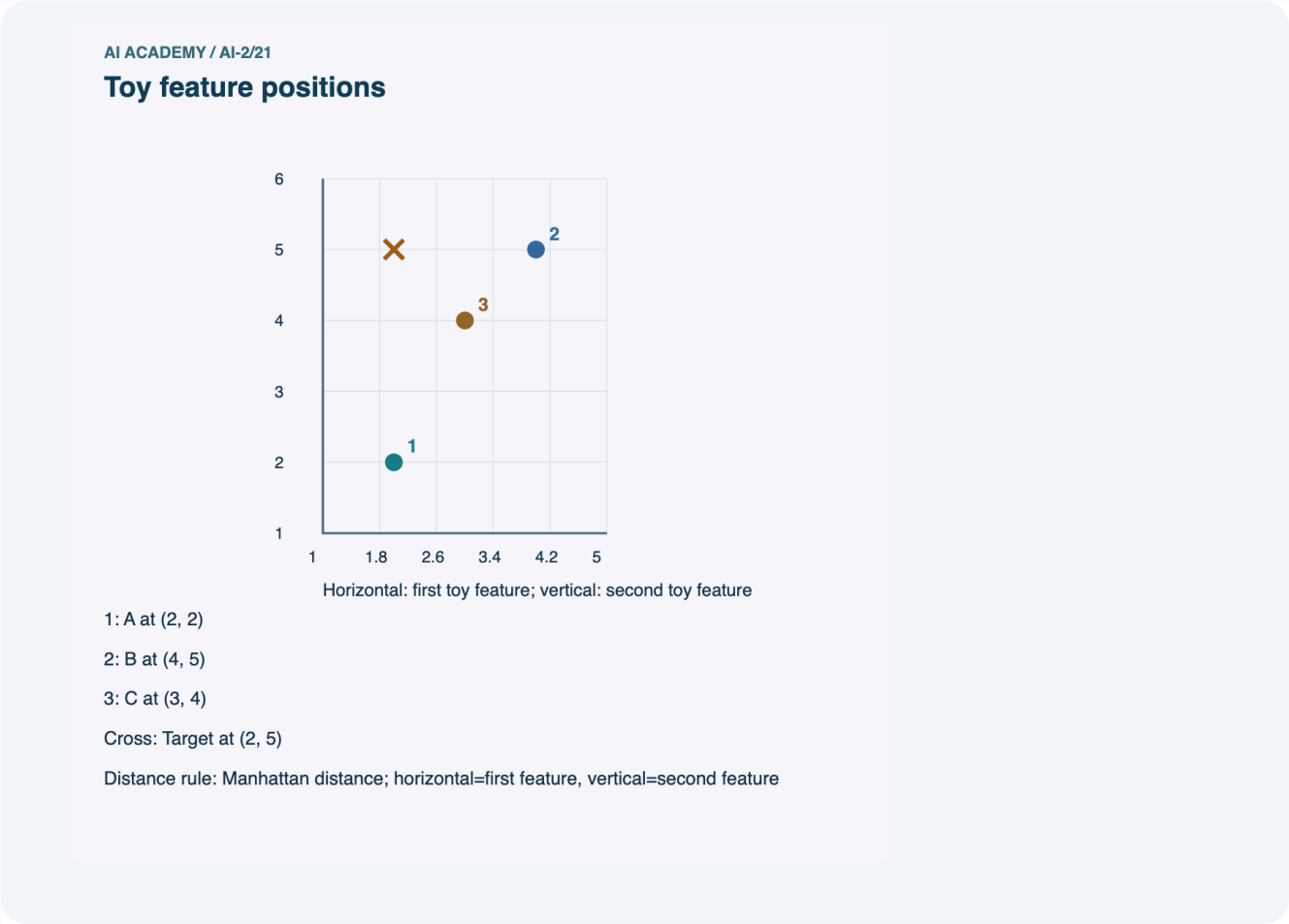

Follow the labelled picture with the matching explanation. Open the vector diagram

**Distance** A: |2−2|+|5−2|=0+3=3. → **Distance** B: |2−4|+|5−5|=2+0=2. → **Distance** C: |2−3|+|5−4|=1+1=2. → Choose closest: B and C tie at **distance** 2; keep the tie visible.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

## 03. Toy feature positions




Follow the labelled picture with the matching explanation. Open the vector diagram

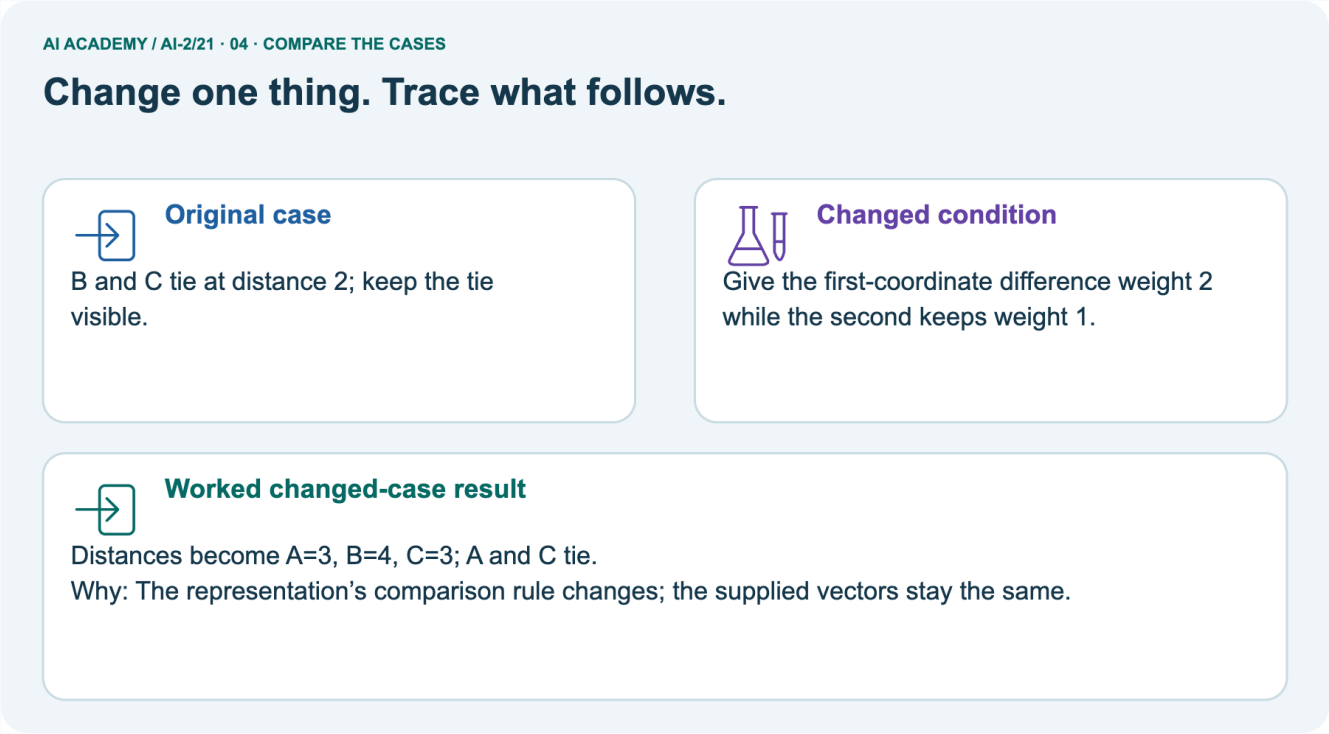
Toy feature positions. A: (2, 2). B: (4, 5). C: (3, 4). Query Target: (2, 5). Axes: first toy feature; second toy feature. **Distance**: **Manhattan distance**; horizontal=first feature, vertical=second feature.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

## 04. Change one thing. Trace what follows.




Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: B and C tie at **distance** 2; keep the tie visible. Changed condition: Give the first-**coordinate** difference weight 2 while the second keeps weight 1. Expected result: **Distance**s become A=3, B=4, C=3; A and C tie.

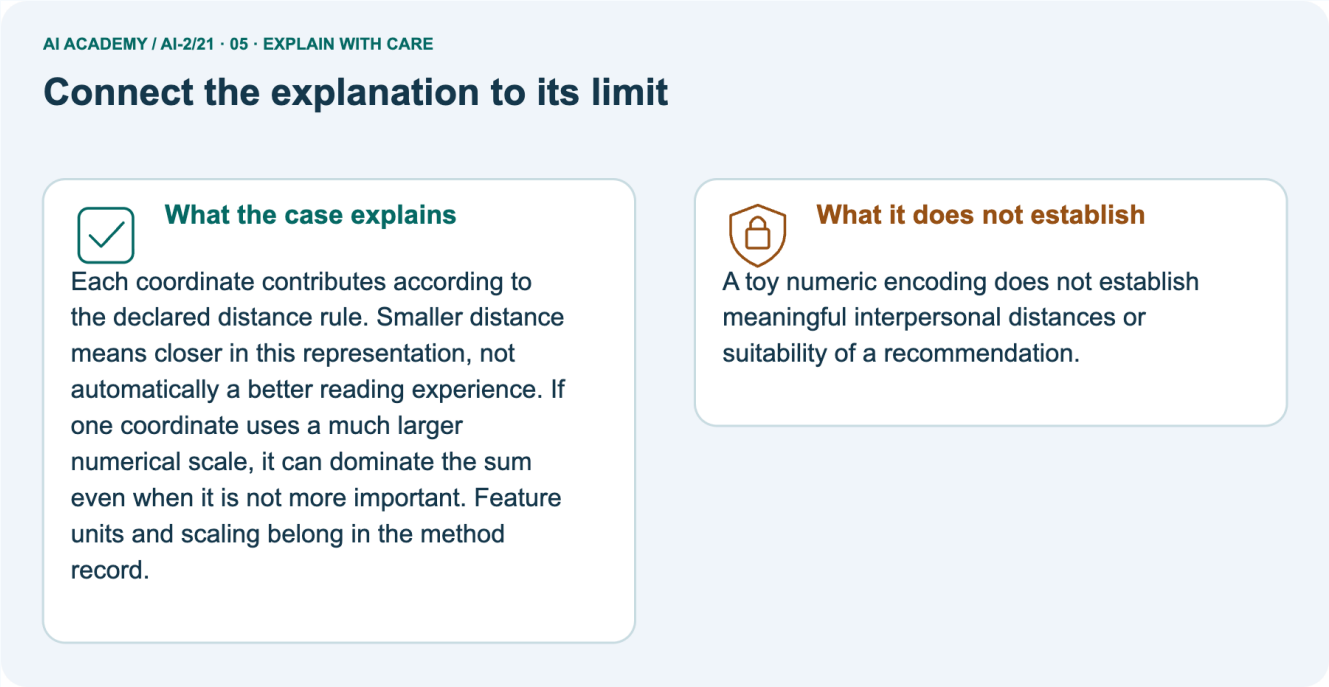
Reason: The representation's comparison rule changes; the supplied vectors stay the same.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The representation's comparison rule changes; the supplied vectors stay the same.

## 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

Each **coordinate** contributes according to the declared **distance** rule. Smaller **distance** means closer in this representation, not automatically a better reading experience.

If one **coordinate** uses a much larger numerical **scale**, it can dominate the sum even when it is not more important. Feature units and scaling belong in the method record.

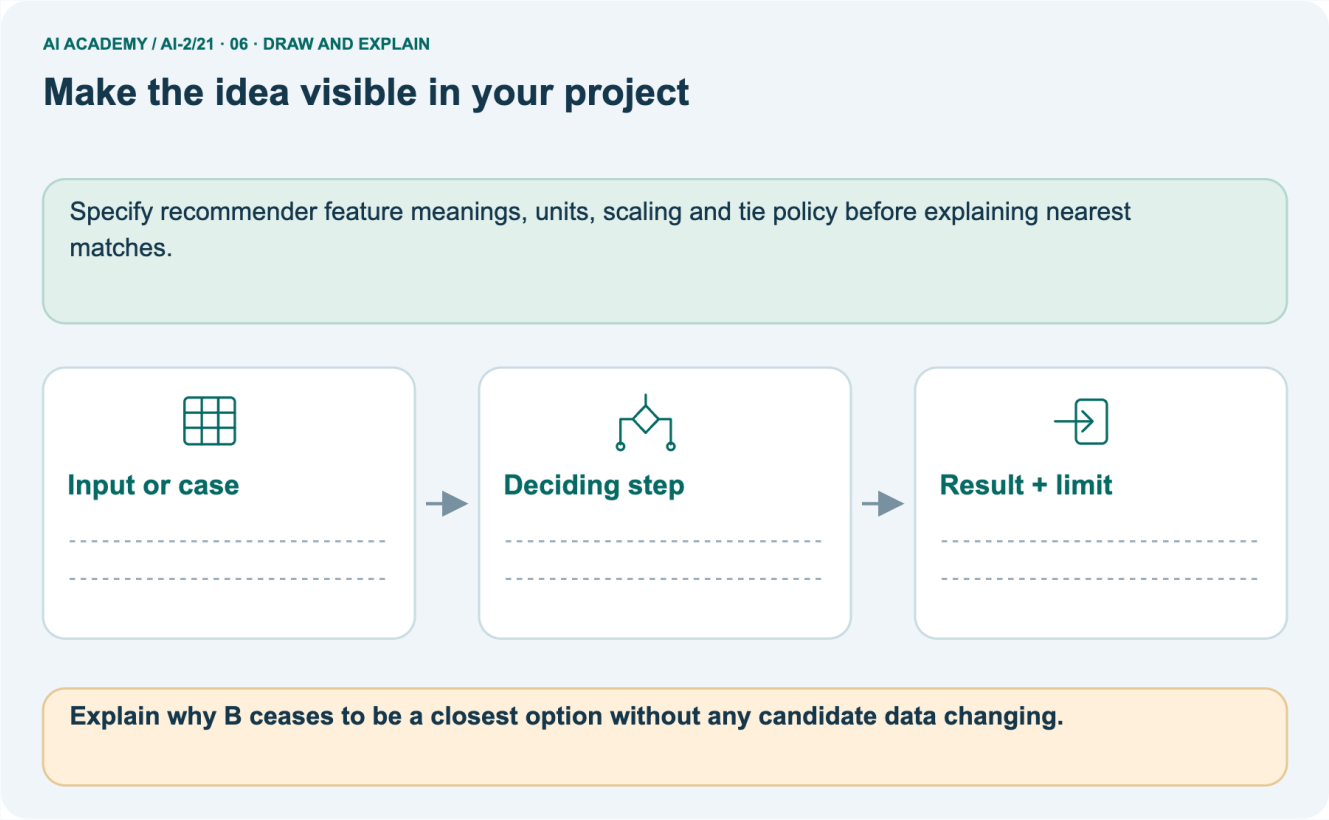
Limit: A toy numeric encoding does not establish meaningful interpersonal **distance**s or suitability of a recommendation.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: A toy numeric encoding does not establish meaningful interpersonal **distance**s or suitability of a recommendation.

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

Specify recommender feature meanings, units, scaling and tie policy before explaining nearest matches. Draw your **input**, deciding step, result and limit.

Supported question: Explain why B ceases to be a closest option without any candidate data changing.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Its first-**coordinate** gap 2 now contributes 4, raising its **distance** above A and C's 3.

## ▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

### 01. Similarity by Selected Traits: the case

### 02. Follow the deciding steps

### 03. Toy feature positions

### 04. Change one thing. Trace what follows.

### 05. Connect the explanation to its limit

### 06. Make the idea visible in your project

## ▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.